Comprehensive Practice Paper: Class 12 Physics (2025-26)

Aligned with Competency-Based Assessment Guidelines

This paper aggregates recurring high-value questions, derivations, and case studies found across previous year papers and sample materials, aligned with the 2025-26 syllabus weightage.

> **General Instructions:**
>
> - **Competency Focus:** 50% of the paper tests conceptual application (Case studies, reasoning).
> - **Format:** Section A (MCQs), B (VSA), C (SA), D (Case-Based), E (Long Answer).

Section A: High-Yield MCQs & Concepts (1 Mark Each)

1. **Electrostatics:** A point P lies at a distance x from the mid-point of an electric dipole on its axis. The electric potential at point P is proportional to:
(a) $1/x^2$ (b) $1/x^3$ (c) $1/x^4$ (d) $1/x^{1/2}$

2. **Current Electricity:** If the potential difference V applied across a conductor is doubled (keeping length constant), how does the drift velocity v_d change?
(a) $v_d/2$ (b) v_d (c) $2v_d$ (d) $4v_d$

3. **Magnetism:** Which of the following statements is **not** true for nuclear forces?
(a) They are stronger than Coulomb forces.
(b) They saturate as the separation between two nucleons increases.
(c) They are always attractive.
(d) They depend on the charge of the nucleons.

4. **EM Waves:** Which electromagnetic waves are used in radar systems?
(a) Infrared waves (b) Ultraviolet rays (c) Microwaves (d) X-rays

5. **Optics:** In a Young's double-slit experiment, the fringe width is β. If the entire apparatus is immersed in a liquid of refractive index μ, the new fringe width will be:
(a) β (b) μβ (c) β/μ (d) β/μ^2

Section B: Conceptual & Very Short Answers (2 Marks)

6. **Current Electricity:** Explain briefly the formation of diffusion current and drift current in a p-n junction diode.

7. **Magnetism:** Define "Current Sensitivity" of a galvanometer. Explain how the current sensitivity of a galvanometer can be increased.

8. **Dual Nature:** The figure shows a plot of V_m^2 (maximum speed squared) versus 1/λ for photoelectrons. Using Einstein's photoelectric equation, obtain the expression for Planck's constant and the work function of the surface.

9. **Atoms:** Calculate the ratio of the nuclear densities of two nuclei having mass numbers 64 and 125.

10. **EM Induction:** Define mutual inductance. Write its SI unit.

Section C: Derivations & Short Answers (3 Marks)

11. **Optics (Wave):** State **Huygens' principle**. Using this principle, draw a ray diagram to show the refraction of a plane wave at a plane surface separating two media and verify **Snell's law** of refraction.

12. **Optics (Ray):** Draw a labelled ray diagram showing the image formation by a **refracting telescope** in normal adjustment. Define its magnifying power. Write two limitations of a refracting telescope over a reflecting telescope.

13. **AC Circuits:** A series LCR circuit is connected to an AC source $v = v_m \sin \omega t$. Derive an expression for the **impedance** of the circuit using a phasor diagram. Find the condition for **resonance**.

14. **Semiconductors:** Explain the working of a **p-n junction diode as a full-wave rectifier** with the help of a circuit diagram. Draw the input and output waveforms.

15. **Nuclei:** Draw a graph showing the variation of **Binding Energy per nucleon** with mass number A. Explain why light nuclei undergo nuclear fusion and heavy nuclei undergo nuclear fission based on this graph.

Section D: Competency & Case-Based Questions (4 Marks)

16. Case Study: Electrostatics (Capacitors)

Context: A parallel plate capacitor consists of two conducting plates separated by a distance. When charged, an electric field is set up.

- (i) Derive the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.
- (ii) If the air between the capacitor plates is replaced by a dielectric of constant K=3, what will be the potential difference between the plates if the charge remains constant?
- (iii) Which physical quantity remains constant when a capacitor is disconnected from a battery and a dielectric slab is inserted?

17. Case Study: Optics (Lenses)

Context: A thin lens is a transparent optical medium bounded by two surfaces. Using the lens maker's formula, one can determine the focal length.

- (i) A biconvex lens (n=1.5) has radii of curvature 20 cm each. Calculate its focal length.
- (ii) If this lens is immersed in water (n=1.33), how does its focal length change?

Section E: Long Answer Derivations (5 Marks)

18. **Electrostatics (Gauss Law):** (a) State **Gauss's Law**. (b) Derive an expression for the electric field due to a **uniformly charged infinite plane sheet**. (c) Find the electric field due to two infinite plane parallel sheets with charge densities +σ and -σ.

19. **Magnetism (Moving Coil Galvanometer):** (a) Write the principle and explain the working with a labelled diagram. (b) Explain why the magnetic field is made radial. (c) How can you convert it into (i) an Ammeter and (ii) a Voltmeter?

20. **Current Electricity:** (a) Two cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel. Derive the expression for the **equivalent emf** and **equivalent internal resistance**. (b) A battery of emf 10 V and internal resistance 3 Ω is connected to a resistor. If the current is 0.5 A, calculate the resistance and the terminal voltage.

Important Topics & Weightage Checklist (2025-26)

Unit	Chapter / Topic	Weightage
Electrostatics	Gauss Law, Capacitance	16 Marks
Current Electricity	Drift Velocity, Kirchhoff's Rules	8 Marks
Magnetism	Biot-Savart, Ampere Law, Galvanometer	17 Marks
AC & EMI	LCR Circuit, Transformer, AC Generator	18 Marks
Optics	Lens Maker’s, Microscope, YDSE, Huygens	17 Marks
Modern Physics	Photoelectric, Bohr's, BE Curve	12 Marks
Semiconductors	V-I Characteristics, Rectifiers	7 Marks

Study Tip: Focus on NCERT diagrams and graphs (especially for Binding Energy and Photoelectric effect) as rote learning is being replaced by competency-based conceptual questions.

Physics Last Paper By Pragya Thunderstudy
(DO NOT REMOVE ANY TEXT)